MIHIKA RAO

Product Analytics | Measurement Strategy

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Astronomy Ph.D. and product analyst focused on what data can tell us, where it falls short, and what to do next. Built measurement and personalization workflows for a learning platform serving 5K+ users; brings Python, SQL, statistical modeling, and clear communication to decisions that need more than a dashboard.

AREAS OF EXPERTISE

Python · SQL · R · Tableau · Product metrics · Cohort and retention analysis · Statistical modeling · Hypothesis testing · Data validation · Research design · Stakeholder communication · Technical writing

PROJECTS

Science Program Evaluation | Dark Skies, Bright Kids *Nov 2021 – Jun 2023*

- Designed pre/post surveys and daily reflections across six after-school clubs to understand whether students left more interested in science, more confident, or both; analyzed responses from 135 students and 540 reflections.
- Used nonparametric tests and ordinal regression in R to examine these outcomes separately and identify activities associated with students feeling like scientists.
- Turned findings into recommendations for creative work, inquiry, and discussion of scientific methods; contributed to a peer-reviewed paper on program outcomes.

Graduate Stipend Analysis | UVA Karsh Institute *Jan 2023 – Oct 2023*

- Built a measurement framework to assess whether graduate stipends covered living costs, combining benchmarks from 25 peer institutions with survey responses from 8K+ students.
- Created executive-facing visualizations that made the funding gap clear and translated the evidence into stipend recommendations for university leadership.
- Analysis contributed to a $10K annual stipend increase affecting 12K+ graduate students.

PROFESSIONAL EXPERIENCE

NovaCards | Education Technology *May 2025 – Aug. 2025*

Data Science Fellow, Product Analytics | Internship *Remote*

- Found that inconsistent and duplicate topic labels made learning performance difficult to measure; prioritized organizing the content before building further analytics for a platform serving 5K+ users.
- Designed a Subject → Topic → Subtopic taxonomy and Python classification pipeline to standardize existing labels and automatically categorize new content.
- Used SQL and Python to identify learning gaps and opportunities for targeted review; defined product metrics and built Tableau dashboards tracking retention, cohort performance, and learning outcomes.
- Reduced recurring data-cleaning and analysis work by 50%+; supported product and personalization changes that contributed to a 30% improvement in 30-day retention during the pilot.

University of Virginia & National Radio Astronomy Observatory *Aug. 2021 – Aug. 2026*

Graduate Research Assistant | Research Appointment *Charlottesville, VA*

- Led 3+ multi-year research projects, turning open-ended questions into analytical plans, testable hypotheses, and validated findings.
- Built end-to-end Python pipelines to clean, validate, and analyze multi-GB datasets, reducing processing time by 40%; created reusable quality checks and visualizations to make repeated analyses consistent and reproducible.
- Applied regression, nonlinear modeling, Monte Carlo simulation, and Bayesian methods to compare competing explanations and determine how strongly the data supported each conclusion.
- Defined questions, validation criteria, and milestones with 20+ collaborators across five institutions; communicated findings through 6+ publications and presentations at six international and ten national conferences.

EDUCATION

University of Virginia *Charlottesville, VA*

Ph.D. Astronomy *Aug. 2026*

M.S. Astronomy | GPA: 3.98 *May 2023*

- **Honors:** Grote Reber Fellowship; Jefferson Foundation Dissertation Fellowship; $375K+ in competitive research support

Agnes Scott College *Decatur, GA*

B.S. Astrophysics · B.A. Economics | GPA: 3.92 *May 2021*

- **Honors:** Summa Cum Laude; Phi Beta Kappa; Presidential Full-Ride Merit Scholarship; NASA and NSF research fellowships